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FAQ Slurm

JOB SUBMISSION

How to submit an interactive job?

You can run an interactive job like this:

```
$ srun --nodes=1 --ntasks-per-node=1 --cpus-per-task=1 --time=01:00:00 --pty  
bash -l
```

Here we ask for a single task, one cpu-core and one interactive node for one hour with the default amount of memory.

Or simply request a bash session using default resources:

```
$ srun --pty bash -l
```

The command prompt will appear as soon as the job starts.

How to request specific node(s) at job submission?

Use **-w** (or **--nodelist**) option, e.g.:

```
$ sbatch -w <node4-1>,<node4-2> ...
```

How to exclude specific nodes from job?

Use **-x** (or **--exclude**) option, e.g.:

```
$ sbatch -x <node4-1>,<node4-2> ...
```

Use -b (or --begin) option in order to defer the allocation of the job until the specified time.

Examples:

```
$ sbatch --begin=20:00 ...           # job can start after 8 p.m.  
$ sbatch --begin=now+1hour ...       # job can start 1 hour after  
submission  
$ sbatch --begin=2023-12-24T20:00:00 ... # job can start after specified date/  
time
```

How to submit dependency (chain) jobs?

Use -d (or --dependency) option, e.g.

```
$ sbatch -d afterany:123456 ...
```

This will defer the submitted job until the specified job `123456` has terminated.

Slurm supports a number of different dependency types, e.g.:

```
-d after:123456      # job can begin execution after the specified job has  
begun execution  
-d afterany:123456   # job can begin execution after the specified job has  
finished  
-d afternotok:123456 # job can begin execution after the specified job has  
failed (exit code not equal zero)  
-d afterok:123456    # job can begin execution after the specified job has  
successfully finished (exit code zero)  
-d singleton         # job can begin execution after any previously job with  
the same job name and user have finished
```



TIP

Note: Multiple jobs can be specified by separating their job ids by colon characters (:), e.g.

```
$ sbatch -d afterany:123456:123457 ...
```

This will defer the submitted job until the specified jobs 123456 and 123457 have both finished.

JOB MONITORING AND CONTROL

How to view information about submitted jobs?

Use **squeue** command, e.g.:

```
$ squeue                                # all jobs owned by user (all jobs owned by all
users for admins)
$ squeue --me                           # all jobs owned by user (same as squeue for regular
users)
$ squeue -u <username>                 # jobs of specific user
$ squeue -t PENDING                    # pending jobs only
$ squeue -t RUNNING                    # running jobs only
```



TIP

The output format of squeue (and most other Slurm commands) is highly configurable to your needs. Look for the `--format` or `--Format` options.

How to cancel jobs?

Use **scancel** command, e.g.

```
$ scancel <jobid>                      # cancel specific job
$ scancel <jobid>_<index>               # cancel indexed job in a job array
$ scancel -u <username>                 # cancel all jobs of specific user
$ scancel -t PENDING                    # cancel pending jobs
$ scancel -t RUNNING                    # cancel running jobs
```

How to get estimated start time of a job?

```
$ squeue --start
```



TIP

Estimated start times are dynamic and can change at any moment.

How to prevent (hold) jobs from being scheduled for execution?

```
$ scontrol hold <job_id>
```

How to unhold job?

```
$ scontrol release <job_id>
```

How to requeue (cancel and resubmit) a particular job?

```
$ scontrol requeue <job_id>
```

How to modify a pending/running job?

Use:

```
$ scontrol update JobId=<jobid> ...
```

E.g.:

```
$ scontrol update JobId=666 TimeLimit=4-0
```

